

Unit 02: Data and Signals

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Objectives

After this lecture, you would be able to:

- understand analog and digital signals
- learn what are the various transmission modes.
- understand what is performance metrics.
- learn the various essential network metrics to monitor.
- Understand the various reasons for transmission impairments
- Understand the concept of protocols.
- Learn the components and functions of protocols

Introduction

One of the major functions of the physical layer is to move data in the form of electromagnetic signals across a transmission medium. Whether you are collecting numerical statistics from another computer, sending animated pictures from a design workstation, or causing a bell to ring at a distant control center, you are working with the transmission of **data** across network connections. Generally, the data usable to a person or application are not in a form that can be transmitted over a network. For example, a photograph must first be changed to a form that transmission media can accept. Transmission media work by conducting energy along a physical path.

2.1 What are major functions of Physical Layer

The physical layer is responsible for electromagnetic compatibility including electromagnetic spectrum frequency allocation and specification of signal strength, analog bandwidth, etc. The transmission medium may be electrical or optical over optical fiber or a wireless IR communication link. For example, a photograph must first be changed to a form that transmission media can accept.

Analog and Digital Data

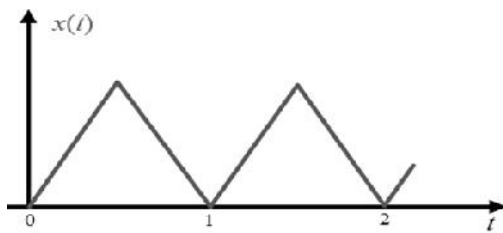
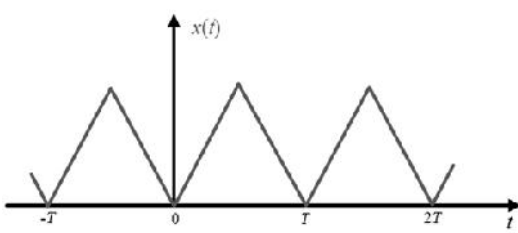
Signals can be classified based on different parameters like the time intervals between which it is being created, or the type of the signal. Both data and the signals that represent them can be either analog or digital in form as shown in Table 1.

Table 1 Analog and Digital Data

Analog Data	Digital Data
The term analog data refers to information that is continuous.	Digital data take on discrete values.

2.2 Signal Classifications

a) Periodic and Non-Periodic Signals

Periodic Signals	Non-Periodic Signals
Completes a pattern and repeats that pattern over subsequent identical periods.	Changes without exhibiting a pattern
The completion of one full pattern is called a cycle.	Any continuous-time signal which is not periodic is called a non-periodic signal.
	

b) Analog and Digital Data

Examples of Analog data are shown below.



An analog clock that has hour, minute, and second hands gives information in a continuous form.



Sounds made by a human voice take on continuous values.



The example of digital data are shown below:



Digital clock that reports the hours and the minutes will change suddenly from 8:05 to 8:06.

i Analog Signal

- Many levels of intensity
- An infinite number of values.

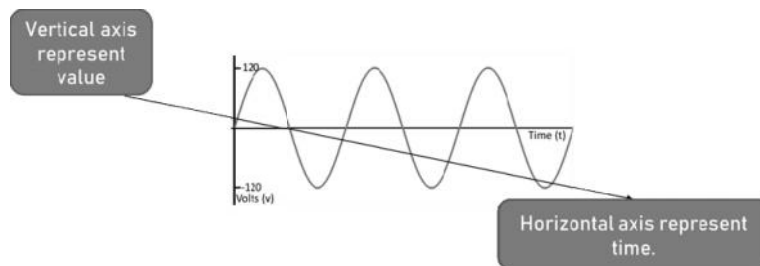


Figure 1 Analog Signal

ii Digital Signal

- Limited number of defined values.
- Although each value can be any number, it is often as simple as 1 and 0.

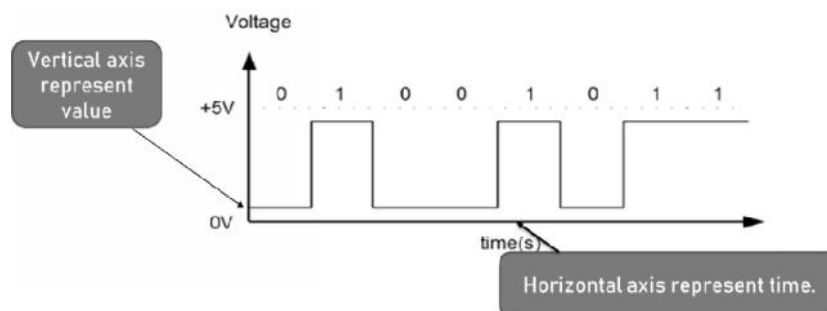


Figure 2 Digital Signal

Information can also be represented by a digital signal. For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage. A digital signal can have more than two levels. In this case, we can use zero voltage. A digital signal can have more than two levels. In this case, we can send more than 1 bit for each level. The following figure shows two signals, one with two levels and the other with four. In general, if a signal has L levels, each level needs $\log_2 L$ bits.

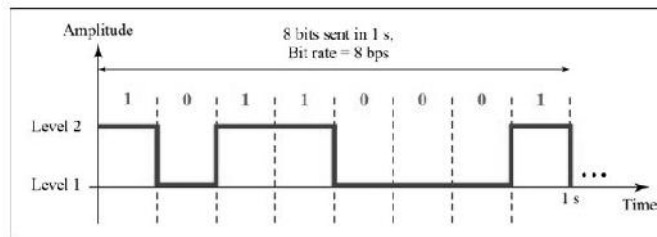


Figure 3 Digital Signal with two levels

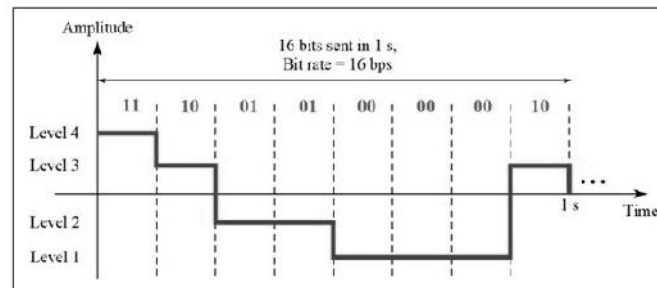


Figure 4 Digital Signal with four levels

Bit Rate:

The bit rate is the number of bits sent in 1s, expressed in bits per second (bps).

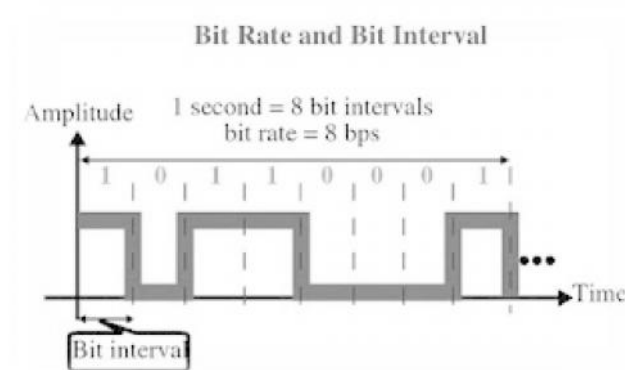


Figure 5 Bit rate and bit interval

The Bit Rate for the above diagram is 8bps and 16bps.

Bit Length:

The bit length is the distance one bit occupies on the transmission medium.

$$\text{Bit length} = \text{propagation speed} \times \text{bit duration}$$

Baud Rate

Baud Rate is the number of signal unit transmitted per second.

Thus Baud Rate is always less than or equal to bit rate. Baud rate is number of symbols per second.

Difference Between Bit Rate and Baud Rate

Bit Rate	Baud Rate
Bit rate is also defined as per second travel number of bits.	Baud rate is also defined as per second number of changes in signal.
Bit rate emphasized on computer efficiency.	While baud rate emphasized on data transmission.
Bit rate is not used to decide the requirement of bandwidth for transmission of signal.	While baud rate is used to decide the requirement of bandwidth for transmission of signal.

Bit Length

The bit length is the distance one bit occupies on the transmission medium.

$$\text{Bit length} = \text{propagation speed} \times \text{bit duration}$$

Bit Interval

Data can be represented by a digital signal. For Example a 1 can be encoded as a positive voltage and a 0 can be encoded as a zero voltage.

2.3 Transmission Mode

Transmission mode refers to the mechanism of transferring of data between two devices connected over a network. It is also called Communication Mode. These modes direct the direction of flow of information.

Modes of Transmission

Transmission mode means transferring of data between two devices. It is also known as communication mode. Buses and networks are designed to allow communication to occur between individual devices that are interconnected. There are three types of transmission mode:-

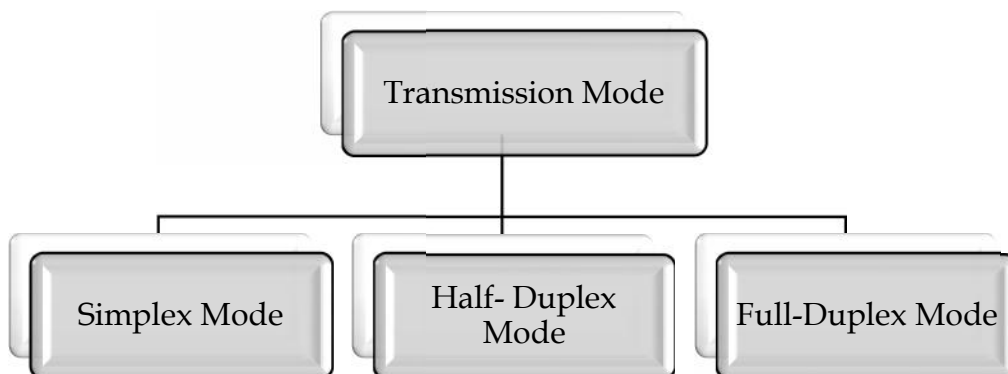


Figure 6 Transmission Modes

These are explained as following below.

a) Simplex Mode -

In Simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit, the other can only receive. The simplex mode can use the entire capacity of the channel to send data in one direction. Example: Keyboard and traditional monitors. The keyboard can only introduce input, the monitor can only give the output.

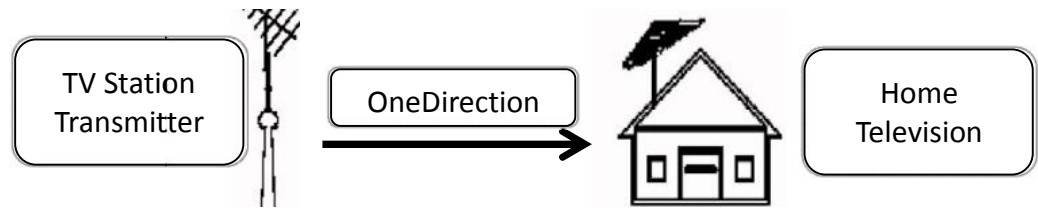


Figure 7 Simplex Mode

b) Half-Duplex Mode -

In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa. The half-duplex mode is used in cases where there is no need for communication in both direction at the same time. The entire capacity of the channel can be utilized for each direction. Example: Walkie- talkie in which message is sent one at a time and messages are sent in both the directions.

$$\text{Channel capacity} = \text{Bandwidth} * \text{Propagation Delay}$$

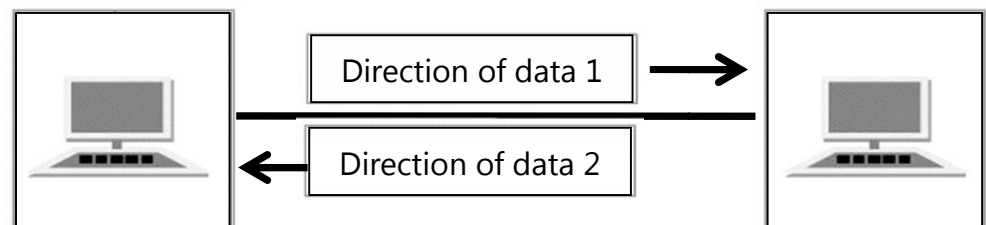


Figure 8 Half Duplex Transmission

c) Full-Duplex Mode -

In full-duplex mode, both stations can transmit and receive simultaneously. In full duplex mode, signals going in one direction share the capacity of the link with signals going in other direction, this sharing can occur in two ways:

Either the link must contain two physically separate transmission paths, one for sending and other for receiving.

Or the capacity is divided between signals travelling in both directions.

Full-duplex mode is used when communication in both direction is required all the time. The capacity of the channel, however, must be divided between the two directions.

Example: Telephone Network in which there is communication between two persons by a telephone line, through which both can talk and listen at the same time.

$$\text{Channel Capacity} = 2 * \text{Bandwidth} * \text{propagation Delay}$$



Figure 9 Full Duplex Transmission

Performance Metrics

Network performance can be affected by a number of different factors. It's important for companies to know which network performance metrics are important to examine. However, depending on the specific issues that plague your network, not every metric is going to be important for you to look at. Despite this, there are some metrics that are essential for any businesses to consider.

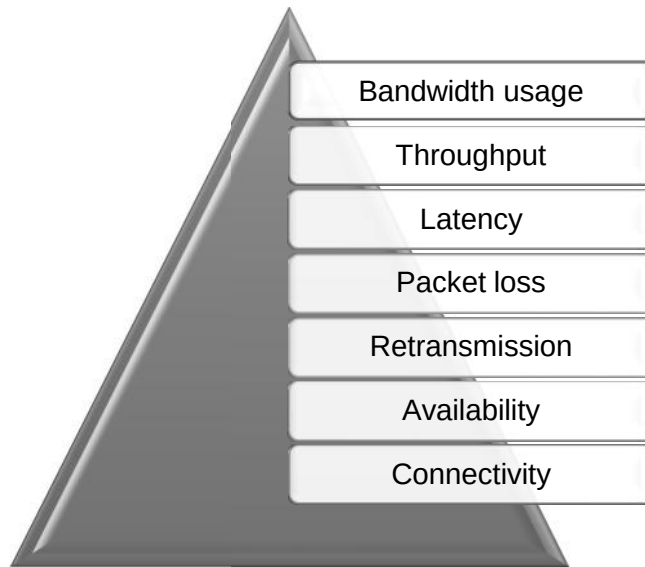


Figure 10: Seven Essential Network Performance Metrics

Bandwidth usage

Bandwidth is the maximum data transmission rate possible on a network. For optimal network operations, you want to get as close to your maximum bandwidth as possible without reaching critical levels. This indicates that your network is sending as much data as it can within a period of time but isn't being overloaded. An NPM can monitor how much bandwidth is currently being used on a network, as well as how much bandwidth is typically used during daily operations. The solution can also alert you when your network is using too much bandwidth.

Throughput

Throughput measures your network's actual data transmission rate, which can vary wildly through different areas of your network. While your network's bandwidth measures the theoretical limit of data transfer, throughput tells you how much data is actually being sent. Specifically, throughput measures the percentage of data packets that are successfully being sent; a low throughput means there are a lot of failed or dropped packets that need to be sent again.

Latency

Latency is the delay that happens between a node or device requesting data and when that data is finished being delivered. This delay can happen for a variety of reasons, but whatever the cause, your NPM solution can track any delays and log them. Consistent delays or odd spikes in delay time indicate a major performance issue; however, because delays can often be undetectable to the human eye, you need a monitoring tool to keep an eye on any delays that happen.

Packet loss

Packet loss examines how many data packets are dropped during data transmissions on your network. The more data packets that are lost, the longer it takes for a data request to be fulfilled. Your IT team should know how many packets are being dropped on average across your infrastructure. A network's Transmission Control Protocol (TCP) interprets when packets are dropped and takes steps to ensure that data packets can still be transmitted; your network team should monitor this system to make sure it's working.

Retransmission

When packets are lost, the network needs to retransmit it to complete a data request. This retransmission rate lets your enterprise know how often packets are being dropped, which is an indication of congestion on your network. You can analyze retransmission delay, or the time it takes for a dropped packet to be retransmitted, to understand how long it takes your network to recover from packet loss.

Availability

Network availability, also known as uptime, simply measures whether or not the network is currently operational. You can never guarantee 100% availability, but you want to be aware of any downtime that happens on your network that you weren't expecting. It's important to be alerted when the network goes down, which network monitoring tools will provide for you. However, you should also be able to discover your actual uptime percentage and how often your network goes down.

Connectivity

Connectivity refers to whether the connections between the nodes on your network are working properly. If there is an improper or malfunctioning connection on your network, it can be a major hurdle for your company. Ideally, every connection should always be operating at peak levels. However, performance issues like malware can target specific nodes or connections to affect performance in that specific area of the network.

2.4 Transmission Impairments

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received.

Reasons for Impairments

Three causes of impairment are attenuation, distortion, and noise.

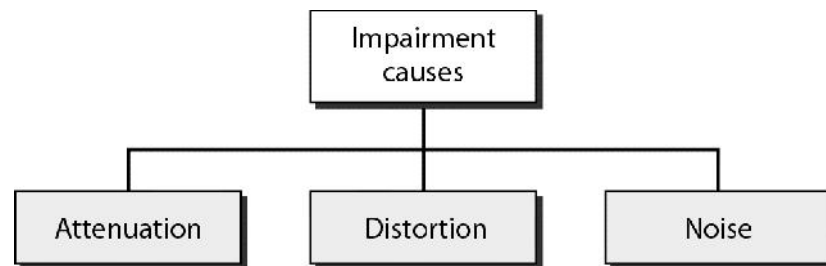


Figure 11 Causes of Impairments

a) Attenuation

It means loss of energy. The strength of signal decreases with increasing distance which causes loss of energy in overcoming resistance of medium. This is also known as attenuated signal. Amplifiers are used to amplify the attenuated signal which gives the original signal back and compensate for this loss.

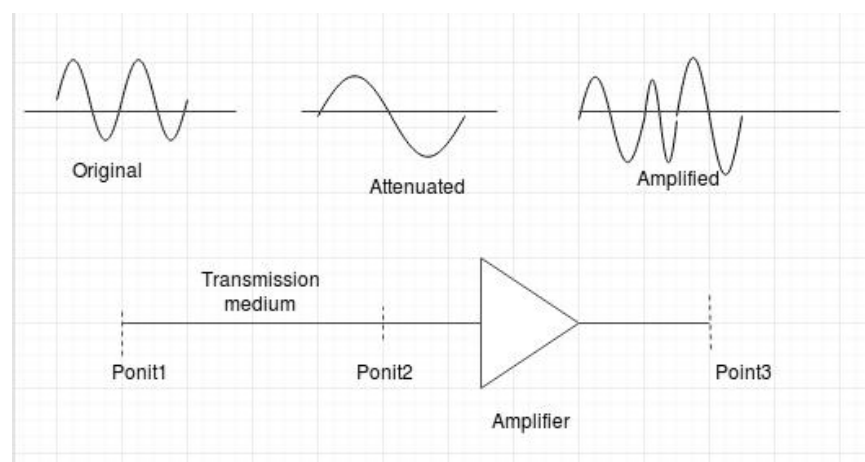


Figure 12 Attenuation

Attenuation is measured in **decibels(dB)**. It measures the relative strengths of two signals or one signal at two different point.

How is Attenuation Measured

To show the loss or gain of energy the unit “decibel” is used.

$$\text{Attenuation dB} = 10\log_{10}P_2/P_1$$

P1 - input signal

P2 - output signal

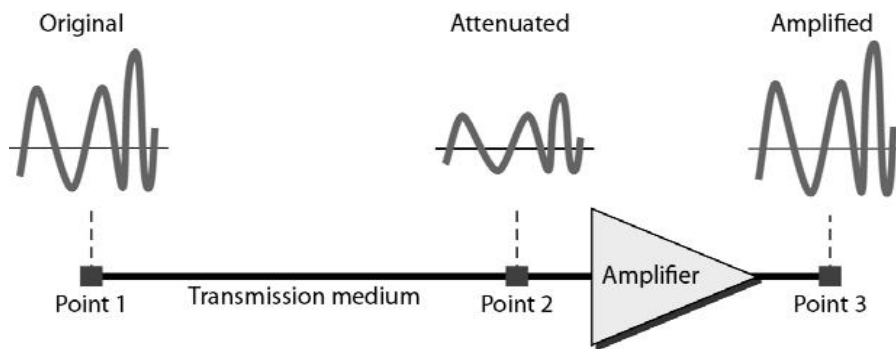


Figure 13 Attenuated Signal



Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that P2 is (1/2)P1. In this case, the attenuation (loss of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5 P_1}{P_1} = 10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}$$

A loss of 3 dB (–3 dB) is equivalent to losing one-half the power



A signal travels through an amplifier, and its power is increased 10 times. This means that P2 = 10P1. In this case, the amplification (gain of power) can be calculated as

$$\begin{aligned} 10 \log_{10} \frac{P_2}{P_1} &= 10 \log_{10} \frac{10P_1}{P_1} \\ &= 10 \log_{10} 10 = 10(1) = 10 \text{ dB} \end{aligned}$$



One reason that engineers use the decibel to measure the changes in the strength of a signal is that decibel numbers can be added (or subtracted) when we are measuring several points (cascading) instead of just two. In Figure 3.27 a signal travels from point 1 to point 4. In this case, the decibel value can be calculated as

$$\text{dB} = -3 + 7 - 3 = +1$$

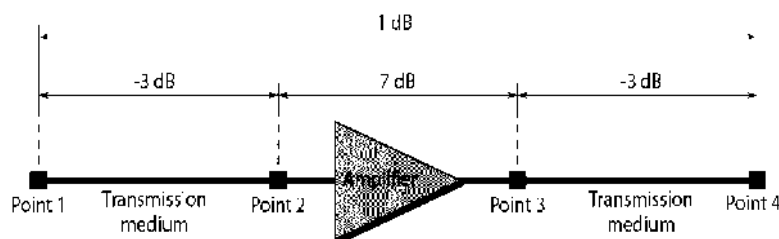


Figure 14 Attenuated Signal over Four Points

b) Distortion

It means changes in the form or shape of the signal. This is generally seen in composite signals made up with different frequencies. Each frequency component has its own propagation speed travelling through a medium. And that's why it delays in arriving at the final destination. Every component arrives at different time which leads to distortion. Therefore, they have different phases at receiver end from what they had at sender's end.

Example

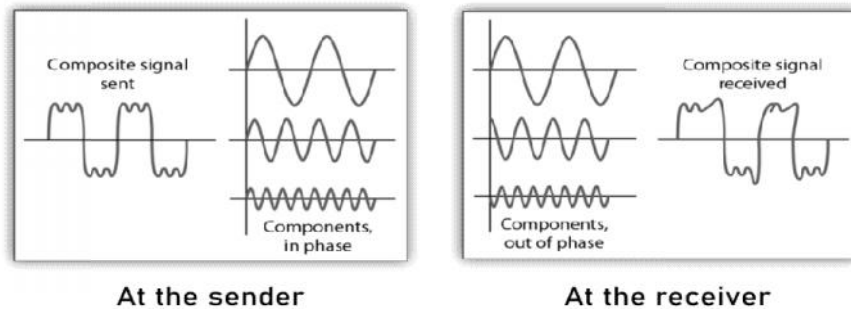


Figure 15 Distortion

c) Noise

The random or unwanted signal that mixes up with the original signal is called noise. There are several types of noise such as induced noise, crosstalk noise, thermal noise and impulse noise which may corrupt the signal.

Induced noise comes from sources such as motors and appliances. These devices act as sending antenna and transmission medium act as receiving antenna. Thermal noise is movement of electrons in wire which creates an extra signal. Crosstalk noise is when one wire affects the other wire. Impulse noise is a signal with high energy that comes from lightning or power lines.

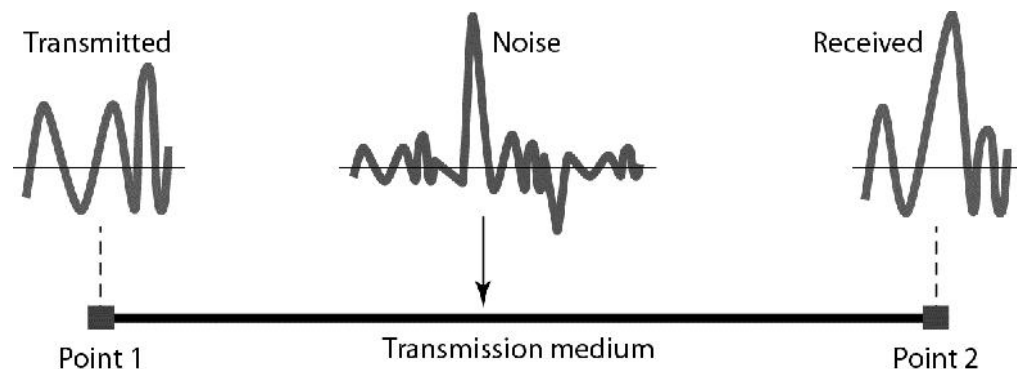


Figure 16 Noise

Signal to Noise Ratio (SNR)

To measure the quality of a system the SNR is often used. It indicates the strength of the signal wrt the noise power in the system.

$$SNR = \frac{\text{Average signal power}}{\text{Average noise power}}$$

It is usually given in dB and referred to as SNR_{dB} .

A high SNR means the signal is less corrupted by noise.

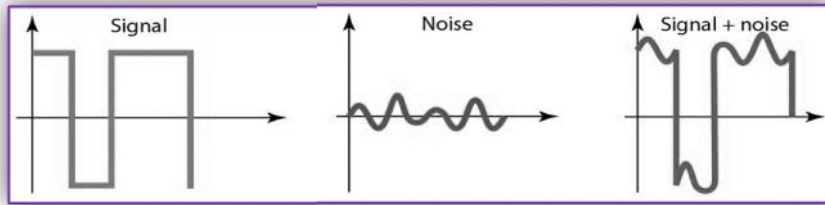


Figure 17 High SNR Signal

- A low SNR means the signal is more corrupted by noise.

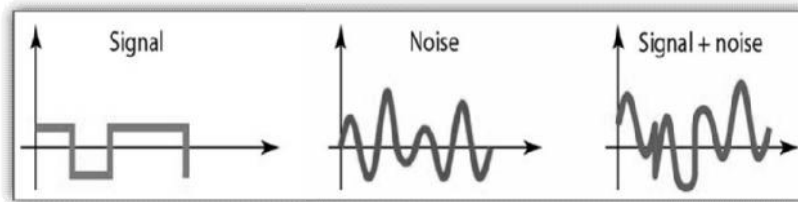


Figure 18 Low SNR Signal

2.5 Protocols

A network protocol is an established set of rules that determine how data is transmitted between different devices in the same network. Essentially, it allows connected devices to communicate with each other, regardless of any differences in their internal processes, structure or design. Cooperative action is necessary. It should be noted that computer networking is not only to exchange bytes. It is a huge system with several utilities and functions. For example error detection, encryption, routing etc. For proper communication, entities in different systems must speak the same language. There must be mutually acceptable conventions and rules about the content, timing and underlying mechanisms. Those conventions and associated rules are referred as "PROTOCOLS".

Protocol Architecture

The task of data transfer is broken up into some modules. It is important to understand as to why it is done and how do these modules interact? For example, file transfer could use three modules: File transfer application, communication service module and network access module. Let us see a real-world example of the Protocol Architecture.

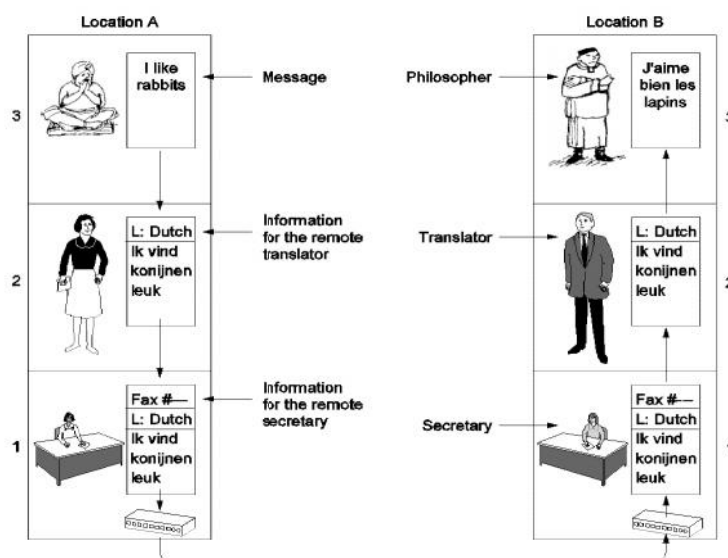


Figure 19 Philosopher-Translator-Secretary Architecture

Let us focus on some of the issues: like the peer-to-peer protocols are independent of each other for example, secretaries may change the comm. medium to email or the translators may agree on using another common language. Note that each layer adds a header

2.6 Network Standards

Networking standards define the rules for data communications that are needed for interoperability of networking technologies and processes. Standards help in creating and maintaining open markets and allow different vendors to compete on the basis of the quality of their products while being compatible with existing market products.

Types of Standards

Standards are of two types i.e. De facto and De jure.

De facto standards are the standards that are followed without any formal plan or approval by any organization. They have come into existence due to traditions or facts. For example, the HTTP had started as a de facto standard.

De jure standards are the standards which have been adopted through legislation by any officially recognized standards organization. Most of the communication standards that are used today are de jure standards.

Standards Organizations

Some of the noted standards organizations are

1. International Standards Organization (ISO)
 2. International Telecommunication Union (ITU)
 3. Institute of Electronics and Electrical Engineers (IEEE)
 4. American National Standards Institute (ANSI)
 5. Internet Research Task Force (IRTF)
 6. Electronic Industries Association (EIA)
 7. World Wide Web Consortium (W3C)
1. **International Standards Organization (ISO)**- The International Organization for standardization widely known as ISO, is an international standard-setting body composed of representatives from various national standards organizations. Founded on February 23, 1947, the organization promulgates worldwide proprietary industrial and commercial standards. It has its headquarters in Geneva, Switzerland. While ISO defines itself as a non-governmental organization, its ability to set standards that often-become law, either through treaties or national standards, makes it more powerful than most non-governmental organizations. In practice, ISO acts as a consortium with strong links to governments. ISO is an international standard-setting body composed of representatives from various national standards organizations the organization promulgates worldwide proprietary industrial and commercial standards. ISO's main products are the International Standards. ISO also publishes Technical Reports, Technical Specifications, Publicly Available Specifications, Technical Corrigenda, and Guides.
 2. **International Telecommunication Union (ITU)** - The International Telecommunication Union is the specialized agency of the United Nations which is responsible for information and communication technologies. ITU coordinates the shared global use of the radio spectrum, promotes international cooperation in assigning satellite orbits, works to improve telecommunication infrastructure in the developing world and establishes worldwide standards. ITU also organizes worldwide and regional exhibitions and forums, such as ITU TELECOM WORLD, bringing together representatives of government and the telecommunications and ICT industry to exchange ideas, knowledge and technology.
 3. **Institute of Electronics and Electrical Engineers (IEEE)** - IEEE's Constitution defines the purposes of the organization as "scientific and educational, directed toward the advancement

of the theory and practice of Electrical, Electronics, Communications and Computer Engineering, as well as Computer Science, the allied branches of engineering and the related arts and sciences." The IEEE is Not-for-Profit Corporation. It was formed in 1963 by the merger of the Institute of Radio Engineers (IRE, founded 1912) and the American Institute of Electrical Engineers (AIEE, founded 1884). It has more than 400,000 members in more than 160 countries, 45% outside the United States. It is also a leading developer of industrial standards having developed over 900 active industry standards in a broad range of disciplines, including electric power and energy, biomedical technology and health care, information technology, information assurance, telecommunications, consumer electronics, transportation, aerospace, and nanotechnology. IEEE develops and participates in educational activities such as accreditation of electrical engineering programs in institutes of higher learning. IEEE serves as a major publisher of scientific journals and a conference organizer. IEEE is one of the leading standards-making organizations in the world. IEEE performs its standards making and maintaining functions through the IEEE Standards Association (IEEE-SA). One of the more notable IEEE standards is the IEEE 802 LAN/MAN group of standards which includes the IEEE 802.3 Ethernet standard and the IEEE 802.11 Wireless Networking standard.

4. **American National Standards Institute (ANSI)** -Though ANSI itself does not develop standards, the Institute oversees the development and use of standards by accrediting the procedures of standards developing organizations. ANSI accreditation signifies that the procedures used by standards developing organizations meet the Institute's requirements for openness, balance, consensus, and due process. ANSI was originally formed in 1918, when five engineering societies and three government agencies founded the American Engineering Standards Committee (AESC). In 1928, the AESC became the American Standards Association (ASA). In 1966, the ASA was reorganized and became the United States of America Standards Institute (USASI). The present name was adopted in 1969. ANSI also designates specific standards as American National Standards, or ANS, when the Institute determines that the standards were developed in an environment that is equitable, accessible and responsive to the requirements of various stakeholders. The American National Standards process involves:
 - consensus by a group that is open to representatives from all interested parties
 - broad-based public review and comment on draft standards
 - consideration of and response to comments
 - incorporation of submitted changes that meet the same consensus requirements into a draft standard
 - availability of an appeal by any participant alleging that these principles were not respected during the standards-development process.
5. **Internet Research Task Force (IRTF)** -The Internet Research Task Force (IRTF) promotes research of importance to the evolution of the Internet by creating focused, long-term Research Groups working on topics related to Internet protocols, applications, architecture, and technology. The IRTF is composed of several focused and long-term Research Groups. Research Groups have the stable long-term membership needed to promote the development of research collaboration and teamwork in exploring research issues. Participation is by individual contributors, rather than by representatives of organizations. The IRTF is managed by the IRTF Chair in consultation with the Internet Research Steering Group (IRSG). The IRSG membership includes the IRTF Chair, the chairs of the various Research Groups, and other individuals ("members at large") from the research community selected by the IRTF Chair. The IRTF is managed by the IRTF Chair in consultation with the Internet Research Steering Group (IRSG). The IRSG membership includes the IRTF Chair, the chairs of the various Research Groups, and other individuals ("members at large") from the research community selected by the IRTF Chair.
6. **World Wide Web Consortium (W3C)** -The World Wide Web Consortium (W3C) is the main international standards organization for World Wide Web (abbreviated WWW or W3). Founded and headed by Tim Berners-Lee, the consortium is made up of member organizations which maintain full-time staff for the purpose of working together in the development of standards for the World Wide Web. W3C also engages in education and outreach, develops software and serves as an open forum for discussion about the Web. W3C was created to ensure compatibility and agreement among industry members in the adoption of new standards. Prior to its creation, incompatible versions of HTML were offered by different vendors, increasing the potential for inconsistency between web pages. The

consortium was created to get all those vendors to agree on a set of core principles and components which would be supported by everyone.

Summary

Having discussed the various transmission medias and the different type of signals that they deal with, a proper transmission mode has to be chosen for the type of communication we are undergoing. To ensure the effective and error-free transmission different essential network performance metrics must be considered to make necessary changes in the network. The different transmission impairments need to be understood by an organisation and should be effectively managed to ensure effective data communication

Keywords

Archive: A computer site advertises and stores a large amount of public domain, shareware software and documentation.

Broadcast Networks: They have a single communication channel, which is shared by all the computers on the network and therefore, any message transmitted by a computer on the network is received by all the computers connected to the channel.

Bit Length: It is the distance one bit occupies on the transmission medium.

Baud Rate is the number of signal unit transmitted per second. It is always less than or equal to bit rate.

Distortion: It means changes in the form or shape of the signal. This is generally seen in composite signals made up with different frequencies. Each frequency component has its own propagation speed travelling through a medium. And that's why it delays in arriving at the final destination

Self-Assessment

1. Telegraph Signals are the examples of
 - a. Digital Signals
 - b. Analog Signals
 - c. Impulse Signals
 - d. Pulse Train
2. The example of an analog to analog conversion is
 - a. Radio
 - b. Video
 - c. Television
 - d. Internet
3. Analog to Analog conversion can be accomplished in
 - a. One way
 - b. Two ways
 - c. Three ways
 - d. Four ways
4. In the _____ transmission mode, communication is unidirectional
 - a. Simplex
 - b. Half-duplex
 - c. Full-duplex
 - d. Hybrid
5. The _____ is an example of simplex device.
 - a. Repeater
 - b. Tap
 - c. Walkie-talkie

- d. Keyboard
6. In the _____ transmission mode, each station can transmit , but not at the same time
- Simplex
 - Half-duplex
 - Full-duplex
 - B and c
7. In the _____ transmission mode, both stations can transmit and receive at the same time.
- Simplex
 - Half-duplex
 - Full-duplex
 - B and c
8. _____ refers to the direction of signal flow between two linked devices
- Line configuration
 - Topology
 - Transmission mode
 - Line discipline.
9. The term that refers to loss of strength of a signal is called
- Attenuation
 - Distortion
 - Noise
 - Impairments
10. A transmission media can have signal impairment because of
- Noise
 - Attenuation
 - Distortion
 - All of above
11. _____ is a type of transmission impairment in which the signal loses strength due to the different propagation speeds of each frequency that makes up the signal.
- Attenuation
 - Noise
 - Distortion
 - Decibel
12. _____ is a type of transmission impairment in which an outside source such as crosstalk corrupts a signal.
- Attenuation
 - Noise
 - Distortion
 - Decibel

Answers for Self Assessment

- | | | | | |
|-------|-------|------|------|-------|
| 1. A | 2. A | 3. C | 4. A | 5. D |
| 6. B | 7. B | 8. C | 9. A | 10. D |
| 11. C | 12. B | | | |

Review Questions

- Q1: Difference between bit rate and baud rate.
- Q2: Explain difference types of modes of communication.
- Q3: Explain difference between full duplex, half duplex and simplex along with examples.
- Q4: Write down the standards of organization.
- Q5: What do you understand by signals. Explain types of signals also.
- Q6: There are always three causes of impairment. Explain them in detail.
- Q7: Difference between periodic and non-periodic signals.
- Q8: Write note on
- a) Bit rate
 - b) Baud rate
 - c) Bit length
 - d) Bit Interval.
- Q9: What do you understand by Defacto and DeJure.
- Q10: What do you understand by protocol. Explain protocol architecture.

Further Readings



Andrew S. Tanenbaum, *Computer Networks*, Prentice Hall

Behrouz A. Forouzan and Sophia Chung Fegan, *Data Communications and Networking*, McGraw-Hill Companies

Burton, Bill, *Remote Access for Cisco Networks*, McGraw-Hill Osborne Media
McGraw-Hill Osborne Media

Rajneesh Agrawal and Bharat Bhushan Tiwari, *Computer Networks and Internet*,
Vikas Publication